Week 6: Packaging and Deployment Readiness

University Knowledge Assistant (RAG) over the Student Resource Book (SRB) 2026 · Group 4

MPSTME, NMIMS · 16 October 2026

Summary

- **Same results as Week 5.** The packaged assistant gives the same answers as the Week 5 notebook on all 182 Week 5 test questions: **92.6%** answer accuracy on Test A and **88.0%** on Test B.
- **No made-up answers.** None of the 20 out-of-scope questions received an answer; every one was refused.
- **The confidence score is useful.** High-confidence answers are **96%** correct and medium-confidence answers **91%**. The score separates right from wrong answers with an AUROC of **0.943** (1.0 would be perfect).
- **Works without internet.** With no API key the assistant still answers, by quoting the matching sentence from the SRB together with its source.

1. Objective of Week 6

Weeks 2–5 built and evaluated the assistant inside a research notebook. The goal of Week 6 was to turn the Week 5 system (v3) into a self-contained assistant that a student-facing app can call, and to confirm that packaging did not change its behaviour. Three questions were checked:

- Does the packaged assistant reproduce the Week 5 answers exactly?
- Can the confidence score that comes with each answer be trusted by the app?
- What happens when the hosted language model is unavailable (no API key or no internet)?

2. What was packaged

The packaged assistant keeps the Week 5 design and settings unchanged. Every setting below was chosen on the development set in Week 5; nothing was re-tuned on test data in Week 6.

Stage	What it does	Setting
Knowledge base	60 rule/policy pages of the SRB, split into 100-word passages with page and section kept	30,916 words
Retrieval	Hybrid keyword (BM25) + semantic (MiniLM) search, then cross-encoder re-ranking	$\beta = 0.6$, $\alpha = 0.7$
Answer generation	LLM answers only from the top passages and cites the passage used	qwen/qwen3.8-27b (Groq), k = 2, temperature 0
Grounding check	An answer with a number not found in the passages is replaced by the verified SRB sentence	min. support 0.5
Refusal	Refuses when retrieval confidence is low or the LLM finds no answer	gate 0.24
Conversation memory	Rewrites follow-up questions into stand-alone questions using the chat history	LLM rewrite

Each reply returns the same fields the app needs: the **answer**, the **source** (SRB part, section and PDF page), a **verbatim evidence quote** from that page, and a **confidence score**. A question the SRB does not cover receives the fixed reply *"I could not find this in the Student Resource Book."*

3. Reproducibility check against Week 5

The packaged assistant was run on every Week 5 test question (Test A: the Week 2 held-out set; Test B: the new blind set) and its answers were compared with the Week 5 notebook outputs. All 182 answers are identical,

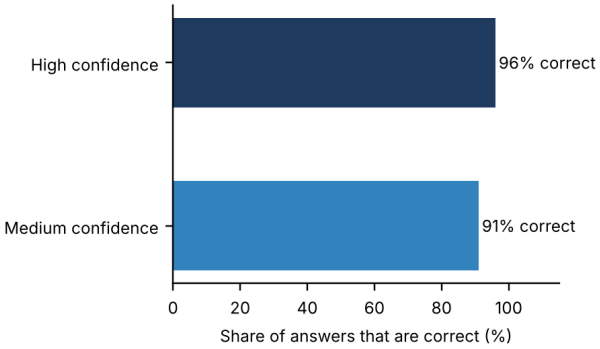
so every Week 5 metric carries over unchanged.

Metric	Test A	Test B	Test A + B
Questions (answerable / out-of-scope)	54 / 6	108 / 14	162 / 20
Answers identical to Week 5	60 / 60	122 / 122	**182 / 182**
Answer accuracy (key fact)	**92.6%**	**88.0%**	89.5%
Citation accuracy (page)	96.3%	93.5%	94.4%
False refusal (answerable questions)	5.6%	5.6%	5.6%
Out-of-scope questions answered	0 / 6	0 / 14	**0 / 20**
End-to-end accuracy (all questions)	93.3%	89.3%	90.7%

Generation is deterministic (temperature 0) and all generated answers are cached, so repeated runs return the same text. End-to-end accuracy counts an answerable question as correct when the key fact is given and an out-of-scope question as correct when it is refused.

4. Is the confidence score trustworthy?

The app can only use the confidence score (for example, to show a “please verify” note) if a higher score really means a more reliable answer. Two checks were made on the Test A + B answers:



Correctness by confidence band. Answers in the high-confidence band are 96% correct, medium-confidence answers 91%. Accuracy falls as confidence falls, which is the behaviour the app needs.

Ranking quality. As a score for separating right from wrong answers, the confidence reaches an **AUROC of 0.943** (1.0 is perfect, 0.5 is chance): a randomly chosen correct case gets a higher score than a randomly chosen wrong one about 94% of the time.

Low-confidence questions are refused rather than answered, so they do not appear as answers.

The confidence score is the mean of three retrieval signals (keyword coverage, semantic similarity and cross-encoder probability), so it can be computed before the LLM is called and is available in offline mode as well.

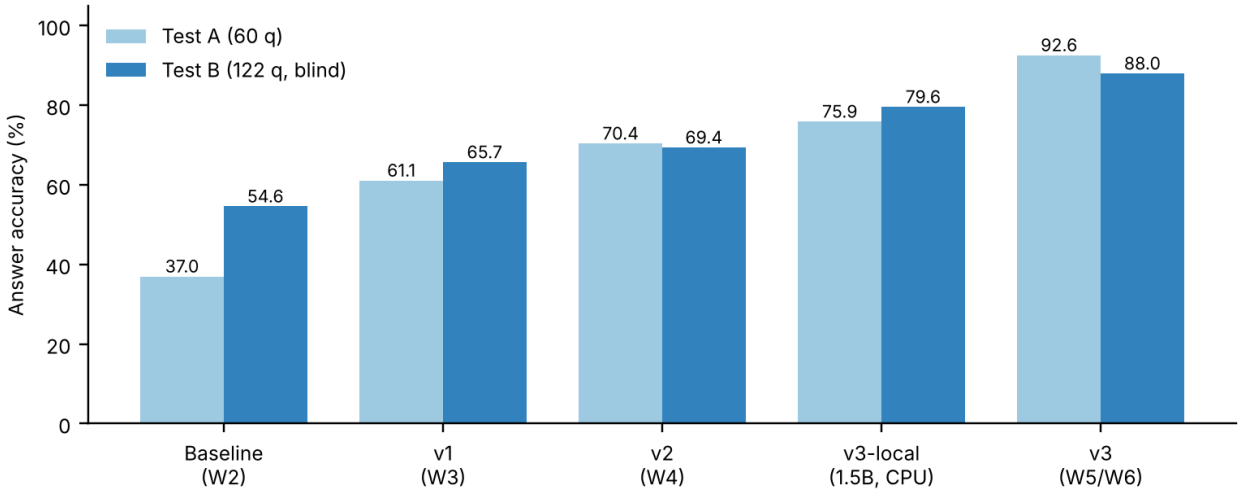
5. Offline mode

If no API key is configured, or the hosted model cannot be reached, the assistant no longer fails. It falls back to the extractive answerer from Week 4: it returns the SRB sentence that best matches the question, copied word for word, with the same source and evidence fields.

	Online (normal)	Offline (fallback)
Answer text	Short answer written by the LLM from the top 2 passages	Best-matching SRB sentence, quoted verbatim
Source and evidence quote	Yes	Yes
Confidence score	Yes	Yes (same retrieval-based score)
Follow-up rewriting	Yes (LLM)	Not available
Needs internet / API key	Yes	No

The quoted sentence cannot contain invented facts, since it is taken directly from the SRB, but it is usually longer and less direct than the LLM answer. In Week 5 the same extractive answerer (v2) reached 69.8% key-fact accuracy on Test A + B, against 89.5% for the LLM answers.

6. Progress across the project



System	Test A acc.	Test B acc.	A + B acc.	Out-of-scope answered
Baseline (Week 2): keyword retrieval + extract	37.0%	54.6%	48.8%	20.0%
v1 (Week 3): hybrid retrieval	61.1%	65.7%	64.2%	25.0%
v2 (Week 4): + cross-encoder re-ranking	70.4%	69.4%	69.8%	25.0%
v3-local: Qwen2.5-1.5B on laptop CPU	75.9%	79.6%	78.4%	25.0%
v3 (Weeks 5–6): RAG chatbot, packaged	**92.6%**	**88.0%**	**89.5%**	**0.0%**

In Week 5 the v3 gain over v2 was statistically significant on Test A + B (33 errors fixed, 1 broken; exact McNemar $p < 0.0001$).

7. Limitations and next steps

- **Retrieval depth.** In 10 of the 17 remaining Week 5 failures the correct passage was ranked 3rd to 5th, just outside the 2 passages the model reads. Reading a third passage is the most direct improvement.
- **Tables and lists in the SRB.** Committee lists, penalty tables and the “People you should know” pages lose their structure in PDF text extraction. Table-aware parsing would fix most of the remaining wrong answers.
- **University vs MPSTME rules.** Where Part I and Part II of the SRB differ, the assistant should prefer the MPSTME (Part II) rule for MPSTME students.
- **Offline refusals.** Without the LLM, refusal relies only on the retrieval-confidence gate, and the LLM's “not found” check is not available. The out-of-scope behaviour of offline mode should be measured on Test A + B before it is relied on, and a stricter gate used if needed.
- **Scale of the test.** The test sets have 182 questions, so one question moves accuracy by 0.5–1.9 points; results should be read with that margin in mind.

Appendix: configuration used

Setting	Value
Generator	qwen/qwen3.8-27b via Groq API (temperature 0); offline: extractive sentence
Embedding model	sentence-transformers/all-MiniLM-L6-v2
Re-ranker	cross-encoder/ms-marco-MiniLM-L-6-v2
Passages read (k)	2
Refusal gate (retrieval confidence)	0.24, no override
Follow-up strategy	LLM rewrite of the follow-up into a stand-alone question
Grounding check	numbers must appear in the passages; content support ≥ 0.5